

ORIGINAL RESEARCH



Development and evaluation of a mobile AAC: a virtual therapist and speech assistant for people with communication disabilities

Erh-Hsuan Wang^{a*}, Leming Zhou^{a*}, Szu-Han Kay Chen^{b†}, Katya Hill^{b*} and Bambang Parmanto^a

^aDepartment of Health Information Management, University of Pittsburgh, School of Health and Rehabilitation Sciences, Pittsburgh, PA, USA;

^bDepartment of Communication Science and Disorders, University of Pittsburgh, School of Health and Rehabilitation Sciences, Pittsburgh, PA, USA

ABSTRACT

Purpose: The currently existing Augmentative and Alternative Communication (AAC) technologies have limitations to produce the best communication rehabilitation outcomes and therefore a better solution is needed.

Method: In this work, a mobile AAC app was developed based on results from research studies. Sophisticated AAC language programming, embedded training materials, and real-time communication performance reporting were integrated into the app. Two groups of study participants were recruited to participate a usability study and a preliminary feasibility study for the purpose of evaluating this mobile AAC app, respectively.

Results: A tablet-based AAC app was developed to support communication rehabilitation. User studies of the app were conducted and included able-bodied individuals and people with verbal communication disabilities. All study participants agreed that the app establishes a usable alternative treatment protocol for communication rehabilitation.

Conclusions: The app's integrated features have great potential to maximize users' communication effectiveness, enhance language skills, and ultimately improve users' quality of life.

ARTICLE HISTORY

Received 27 February 2017

Accepted 16 August 2017

KEYWORDS

mHealth; telehealth; usability study; augmentative and alternative communication; communication disabilities

► IMPLICATIONS FOR REHABILITATION

- We have developed and evaluated an integrated mobile AAC language-based app.
- This tablet-based app integrated AAC with embedded trainings and real-time performance report.

Introduction

Verbal communication is an essential skill for human beings and a major factor influencing quality of life. In the United States, there are over 3.7 million people suffering from profound verbal communication disabilities (such as aphasia and oral-motor speech disorders) [1–3]. People with communication disabilities (PwCD) may benefit from Augmentative and Alternative Communication (AAC) technologies, which can be used to enhance or replace human verbal communication. These AAC strategies and devices enable PwCD to express themselves in spoken language, increase social interactions and maintain independent communication in their everyday life [4]. Based on our clinical experiences, observations of other SLPs and feedback from PwCD — as well as a comprehensive literature review — the extant AAC systems present a number of limitations for PwCD/SLPs to receive/provide effective AAC services and to achieve desired communication outcomes. The following are three major limitations.

First, existing mobile AAC apps can improve communication skills for PwCD. Some of the current designs however may not lead to long-term benefits of language improvement or spontaneous novel utterance generation [5,6]. For example, use of pre-stored sentences may be convenient for PwCD to use the app at first

encounter, however, a previous study on one AAC mobile app showed that AAC users' real-time communication did not include significant use of the pre-stored sentences available in the app [7]. Mainly relying on pre-stored sentences can limit the PwCD's potential to generate user specific sentences to express unique ideas and needs. Ultimately, these AAC apps limit the linguistic and communication competence the PwCD can potentially achieve.

Second, the currently available AAC systems cannot support the recommended intensive treatment at a low cost. Proper intensity of treatment is one of the key components for successful communication rehabilitation across different communication disorders such as aphasia (8 intervention h/week) and autism (38 intervention h/week) [8,9]. Typical insurance plans, however, may cover only 60 h across all types of rehabilitation services each year. In other words, a tremendous gap exists between professional recommendations for language therapy and insurance companies' limits for coverage. The extant AAC systems cannot fill this gap because clinician cannot monitor the PwCD's performance at home, which is related to the third limitation.

Third, the language performance reports from AAC devices are either not conveniently available or do not provide user friendly information. Progress monitoring is one important factor

CONTACT Erh-Hsuan Wang  erw39@pitt.edu  Department of Health Information Management, University of Pittsburgh, School of Health & Rehabilitation Sciences, 6051 Forbes Tower, Pittsburgh, PA, USA

*Chen and Wang contributed equally in developing the application, EuTalk. Chen was the content/conceptual designer, and Wang was the system/interface designer. Hill provided guidance on the contents of the application and Zhou served as a mentor in the design and implementation of the application.

†Present affiliation: Department of Communication Disorders and Sciences, State University of New York at Fredonia Fredonia, NY, USA

in self-regulated task engagement [8]. The feedback provided to PwCD throughout the learning process can be critical to PwCD's engagement in therapeutic tasks [10]. An evidence-based practice also requires clinicians to collect and evaluate PwCD's language performance data so that they can make optimal clinical decisions. Some traditional AAC devices include data logging functions such as language activity monitoring (LAM) to record the usage of AAC devices [11]. Analysing these log files however, is a separate process that does not allow PwCD to obtain real-time feedback about their own language performance when they use traditional AAC devices [12]. Clinicians cannot access the result of log file analysis when they see their patients. A few mobile AAC apps provide outcome displays to their users [13], but it should be noted that the reliability and validity of such outcome data are highly questionable, because their calculation methods may not be well tested or may be based on inadequate standards.

Health information technologies (HITs), especially mobile health (mHealth) technologies developed in recent years, may be used to collect the desired clinical data and make them easily accessible to PwCD, their caregivers, and the clinicians (Speech Language Pathologists or SLPs). mHealth platforms may also be used to deliver healthcare service to fulfill the intensity requirement of language rehabilitation programs. Here, mHealth refers to the use of mobile devices in healthcare with the aim to collect objective patient data in real time and improve the quality of healthcare. Many studies have evaluated the use of mobile apps for supporting health data collection and management, such as disease diagnosis and assessment [14,15], disease management and monitoring [16,17], data collection and reporting [18,19] and health education or promotion [20,21]. All these studies suggest great potential benefits in using mobile devices to provide better quality of patient data and healthcare services, better health service accessibility and also reduce healthcare cost.

Therefore, in this project, mHealth technologies are utilized to improve the patient data collection in the language rehabilitation for PwCD. More specifically, a tablet-based communication app (EuTalk™) for PwCD was created by combining mHealth and AAC technologies to support language performance data collection, reporting and AAC clinical service delivery. Both usability and feasibility studies were conducted to discover design problems and evaluate the usability of the app.

Methods

To address current problems in AAC intervention, a mobile AAC app was designed, implemented and evaluated. The details of these steps are described in the following sections.

Design

The aim of the design is to (1) create an alternative communication solution; (2) follow research-based development guidelines to fully support communication trainings by providing embedded training materials; and (3) generate real-time outcome reports to enhance clinical evaluation and improve user's engagement.

Figure 1 illustrates the key components of this tablet-based mobile AAC app: a communication interface embedded training lessons and performance reports. The communication interface provides both a communication program and clinical treatment exercises. PwCD can use the mobile app to communicate with other people and attend training with/without the guidance from SLPs. The embedded training materials are designed to provide sufficient practice time at home recommended by research in language rehabilitation. All user activities are logged and stored on the tablet and can also be forwarded to a secure server. Outcome reports and language performance measures are generated automatically on the tablet and are accessible to the PwCD and their caregivers, and also their SLPs when they have face-to-face meetings. The outcome reports may help SLPs to determine the result of AAC intervention as well as to make any necessary adjustments to the treatment plan of each patient.

Development

This tablet-based communication app, EuTalk™, was developed on the Android platform and tested on a "Samsung Galaxy Tab 10.1" [22,23]. The VoiceText™ Embedded SDK for Android platform was used to provide the text-to-speech (TTS) feature. The following sections delineate the communication interface, training materials, data collection and reporting and app customization.

Communication interface

The app's communication interface provides the primary language components with graphic symbols that can be implemented in the lexicon to generate desired sentences. For the purposes of

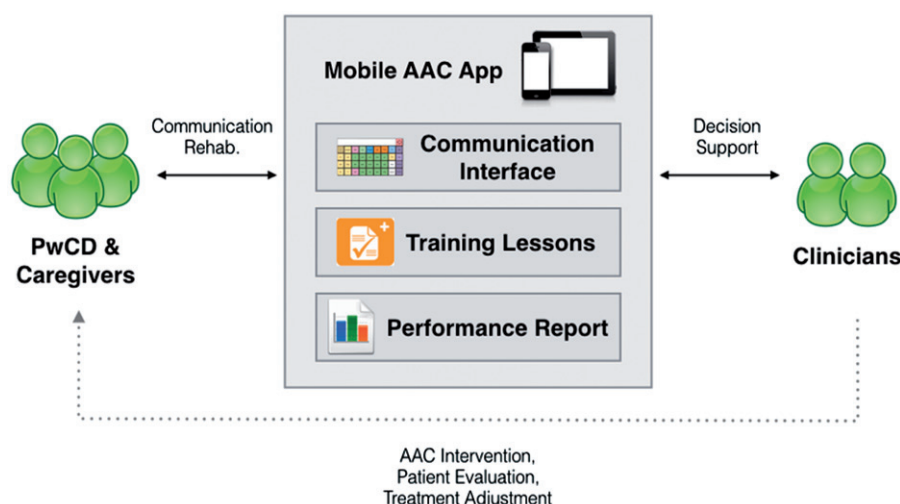


Figure 1. mHealth communication rehabilitation architecture.

this study, a lexicon was selected from word frequency studies that identified core/high frequency words in various contexts [12]. The interface design was based on a previous study [24]. The vocabulary is organized in a 8×4 grid [1], an optimal arrangement given the size of the chosen tablet and the symbols, while one long bar (talk bar) above this grid is used to show the generated sentence. The five functional buttons located at the right hand side of the grid (Figure 2) — “History”, “ABC”, “Pages”, “Favorites” and “Main Menu” — are designed to provide more language elements and customize the settings in the app. For instance, a user might click the ABC button so that he or she can type their sentences letter-by-letter. If the History button is clicked, several recently generated sentences will be shown and can be reused by a single click. The customization settings offered in Pages and Favourites will be introduced in the Customization section.

The mobile app contains 17 pages of this grid type, each linked with words on the main page. Each cell of the grid itself consists of a word and its corresponding symbol taken from the

Pixon™ Project Kit [25]. The customization settings offered in Pages and Favourites will be introduced in the Customization section.

The talk bar on the top allows PwCD to view the utterances they generate, and enables the device to create speech based on the utterances. A delete button, included right after the talk bar, can be used to remove words (one per click) from the generated utterance in the talk bar. Core vocabulary cells in the grid are color-coded based on the word type (pronoun, verb, adjective, etc.). PwCD can construct utterances by simply pressing the words they need. The first page contains the essential vocabulary for sentence construction. Each word on the main page is linked to a connected page with related vocabulary (Figure 3).

Embedded training materials. Because providing a sufficient intensity of treatment and sufficient practice is a key factor to improve/recover PwCD’s language and communication skills, the app was designed to include training lessons to expand regular training so that PwCD can improve their communication skills at home. The embedded training materials are divided into two parts: games and practice lessons.

Previous studies have indicated that applying gamification to a non-game application can improve overall user experience and user engagement [26]. The gamification feature seeks to boost users’ motivation to have more language practice on the app. The training game can help PwCD to become familiar with the main page of the communication interface so that they can quickly find desired words in the page. In the training game, four word sets are provided with different combinations of vocabulary in the main page. Moreover, different levels of difficulty (size of word set and time span of word appearance) are implemented to help PwCD to remember the locations of the main page vocabulary.

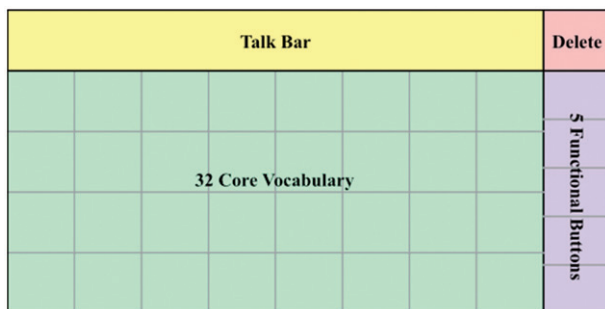


Figure 2. Design of communication interface.



Figure 3. Connected page of communication interface.

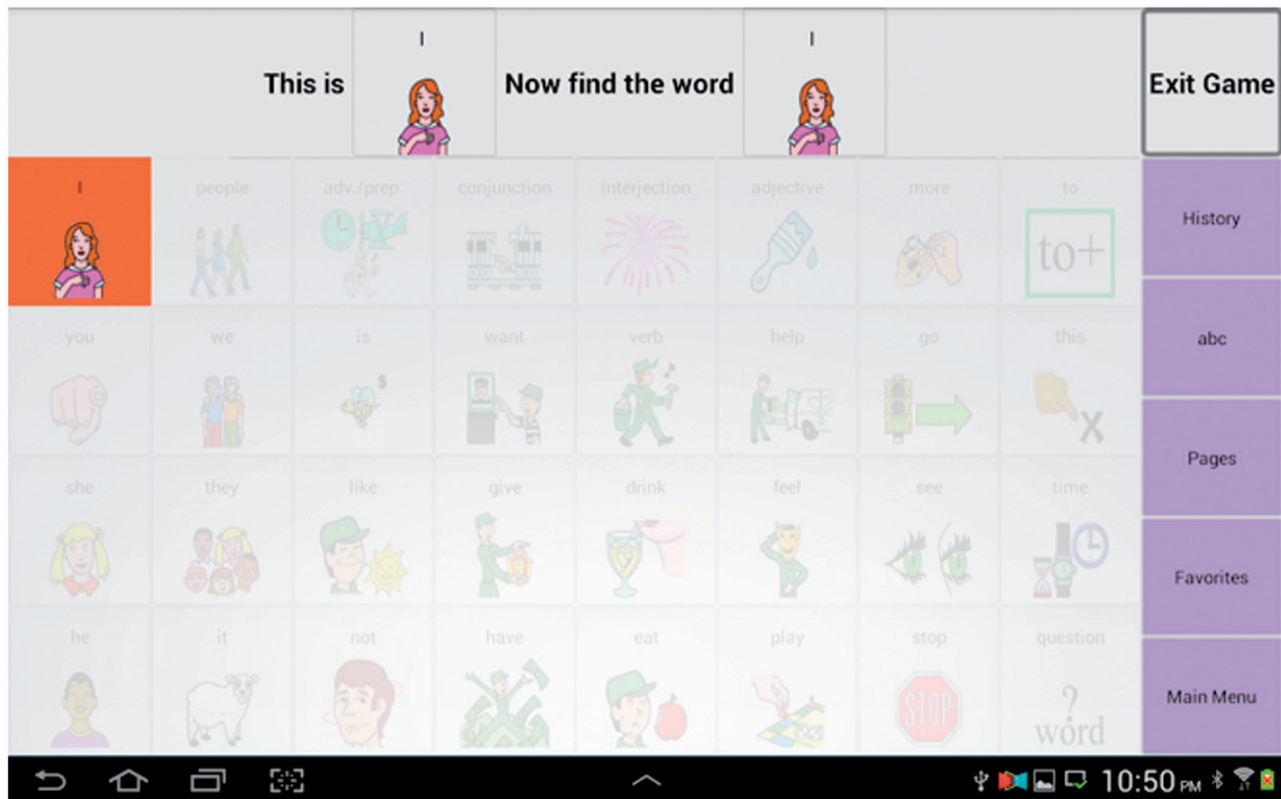


Figure 4. Embedded training – practice.

The practice in the app covers words that are used frequently in daily conversations on the communication interface. The practice tasks have different difficulty levels. The app also provides audio cues (TTS), text and symbols to facilitate the learning process (Figure 4) [27].

Data collection and reporting. Data collection and information delivery is one important feature of the app. This feature was designed to collect user activity data and to automatically analyse the data in real time. The app has a SQLite database to store app information and user generated data.

EuTalk records all user activities and logs all data elements needed by SLPs, based on research study results [4,28]. The databases store users' every click event with a corresponding time-stamp in the app from logging in to logging out.

A report or summary (Figure 5) can be generated in real time based on user-generated data. Six measures of language performance are provided in the AAC app. Records in games and practices are presented as line charts. PwCD can easily see their progress in the training lessons over a period of time. Besides the training results, four more communication measures are provided [11], including communication rate (words per minute), number of individual words, word types and average length of utterances. Line charts are used for both communication rate and average length of utterances since they can show the trend in both measures. The number of individual words is presented in a bar chart to show comparisons over a given time period. A pie chart is used to illustrate numerical proportion of each word type.

Customization. The default settings of the app support basic communication needs. In order to increase the flexibility of the app, customization features in the app can be used. All pages with words linked to the ones in the home page can be customized.

PwCD or their caregivers can add or remove words, according to their personal preferences. SLPs can hide any words on the communication interface to customize the training that they assign to their patients. In the TTS engine, both male and female voices are supported. PwCD or their caregivers can select their preferred voice. "Favourites" is a feature that helps users to store up to 20 sentences that they use often or those that are needed in emergency. For instance, PwCD can store sentences such as their home address, name and contact information of their caregivers, or frequently requested medical information and healthcare providers. Users or their caregivers can type in these sentences using a soft keyboard provided in the app. They can also conveniently update these sentences when necessary.

Evaluation on the app

Two studies were conducted to evaluate the design and development of the app, including one usability study with able-bodied participants, and one feasibility study with PwCD. Both usability and feasibility studies were approved by the University of Pittsburgh Institutional Review Board (IRB). Below are the details of study participant recruitment and the procedure of the two studies.

Study participant recruitment

Able-bodied individuals were recruited for the usability study via recruitment flyers posted in the Greater Pittsburgh area. Participants were healthy native English speakers between the ages of 18 and 65. Additionally, eligible participants must be able to follow oral instruction, operate a tablet device and have no communication disabilities. Twenty (20) adults who met the selection criteria were identified from a pool of volunteers and their eligibility was verified before they participated in this study.



Figure 5. Performance report.

After the problems identified in the able-bodied usability study were fixed, PwCD were recruited for the preliminary feasibility study. The only difference between these participants and the ones in the usability study is that participants in this feasibility study should have communication disabilities. Four adults with communication disabilities who met the selection criteria were identified from a pool of volunteers recruited via clinician referral and the CTSI Research Participant Registry at the University of Pittsburgh. All participants were asked to take the Western Aphasia Battery (WAB) test and use the "Mole Hammer - Whack a Mole" app on an Android tablet to verify their eligibility. This screening procedure was performed before the start of the feasibility study.

Study procedure and tasks

The study procedure and tasks were almost identical in both studies. The experiment consisted of a demographic questionnaire, an introduction to the study, a demonstration of the app, a task section for features in three major components, a self-exploration section, after scenario questionnaires (ASQ), one usability questionnaire and a post-session interview. The tasks on reporting and customization were skipped because of the limited number of data items generated by the study participants. The post session interview was also skipped in the feasibility study because of the communication disabilities of participants. The entire study with each participant took 1.5 to 2 h to complete.

First, a participant was asked to complete a questionnaire on their demographic information. Thereafter the participant was introduced to the mobile app, and was provided a brief demonstration of the app running on the tablet. When the participant was using the app, the think aloud approach was used to collect the participant's feedback on the app. The participant was asked to perform a number of tasks, including using the communication

interface to have a conversation during given scenarios, doing language exercises in the game and practice modes, reading and interpreting the language performance reports, and using the ICU and Favourites panels to communicate with the investigator.

Upon completion of assigned tasks after each session, the participant was asked to complete a post-task questionnaire to collect immediate feedback about the specific features of the app. At the end of the study, the participant was asked to complete a usability questionnaire to provide their overall impression on the app. Furthermore, for able-bodied participants, an interview with open-ended questions was conducted at the end of the study. During the study, all user activity data were logged in the app. The responses to the questionnaires were collected through the Qualtrics system subscribed to by the University of Pittsburgh.

Results

Participants

A total of 20 able-bodied participants participated in the usability study. The participants' ages were between 20 and 64 (mean = 31.4, SD = 14.1); 45% of participants were male ($n = 9$) and 55% of them were female ($n = 11$). Some participants had knowledge of communication sciences and AAC. They all had experience in using mobile devices and most of them typically use mobile devices for at least a few hours each day. Table 1 lists background characteristics and overall AAC literacy of these study participants.

Usability study results

The usability of the app was measured by the modified ASQ and Post Study System Usability Questionnaire (PSSUQ). The ASQs were used to measure the usability of specific features of the app while the PSSUQ was adopted to measure the app's overall

Table 1. Background characteristics of usability study participants.

Gender	N	%
Male	9	45%
Female	11	55%
Education		
High school	10	50%
Bachelor's	6	30%
Master's	4	20%
Age		
18–30	13	65%
31–45	4	20%
45–65	3	15%
Time spent on mobile devices each day		
<1 h	2	10%
1–3 h	13	65%
3–5 h	2	10%
>5 h	3	15%
Mobile devices experience		
iOS only	2	10%
Android only	6	30%
Both iOS and android	11	55%
Amazon kindle	1	5%
Familiar with communication disabilities		
Yes	15	75%
No	5	25%
Familiar with AAC		
Yes	8	40%
No	12	60%

usability [29]. The statements were from ASQ and PSSUQ, but were slightly modified. For instance, the questions would be more relevant when the name of the feature/app was used in the statements. Participants responded to all statements on a 5-point Likert scale that ranged from 1 (strongly disagree) to 5 (strongly agree). In other words, higher numbers indicate a greater level of satisfaction with the app. All 20 participants completed all ASQs and PSSUQ. Table 2 shows the results of these usability assessments. The average of the responses to each statement and the overall average for each major component are listed in Table 2.

Feasibility study results

A total of four participants were recruited to participate in this feasibility study. All four participants were able to complete the assigned tasks and provide feedback through the questionnaires. Participants' ages were 23, 45, 49 and 57 (mean =43.8, SD =14.6); 75% of participants were female ($n = 3$) and 25% were male ($n = 1$). One of the four participants used dedicated AAC devices before.

Four participants completed two ASQ and a modified version of PSSUQ. Table 3 shows usability assessment results from these questionnaires.

Overall performance

The language performance of able-bodied participants on the app was evaluated by analysing the log files. In the usability study,

Table 2. The usability study results.

Statements for communication interface	Mean	SD
1.1 I would find the communication interface easy to use.	3.75	0.64
1.2 I was able to complete the tasks and scenarios using the communication interface.	3.90	0.45
1.3 I liked using the interface of the communication interface.	3.90	0.64
1.4 I would find the communication interface intuitive to use.	3.45	0.83
1.5 I would find the page navigation easy to understand.	3.65	0.75
1.6 I would find the function keys helpful using the communication interface.	3.90	0.55
1.7 I would find the vocabulary provided is sufficient to make daily conversation.	4.10	0.45
Average on communication interface	3.81	–
Statements for embedded training		
2.1 I would find the game easy to use.	4.10	0.72
2.2 I was able to complete the tasks using the game.	4.10	0.72
2.3 I liked using the interface of the game.	4.10	0.55
2.4 I would find the game intuitive to use.	4.00	0.79
2.5 I would find the game helpful to use the communication interface.	4.35	0.67
2.6 I would find the practice easy to use.	4.10	0.64
2.7 I was able to complete the tasks using the practice.	3.90	0.64
2.8 I liked using the interface of the practice.	3.90	0.64
2.9 I would find the practice intuitive to use.	3.74	0.81
2.10 I would find the practice helpful to use the communication interface.	4.21	0.79
2.11 I would find the achievement banner (practice result) motivational.	3.70	1.03
Average on embedded training	4.02	–
Statements for reporting and customization		
3.1 I was able to complete the scenario using the report.	4.00	0.65
3.2 I liked the interface of the report.	4.10	0.55
3.3 I would find the information provided in report easy to understand.	4.25	0.55
3.4 I would find the information provided in report helpful.	3.90	0.72
3.5 I would find the report helpful to use the communication interface.	4.30	0.66
3.6 I was able to complete the scenario using favourites.	3.90	0.72
3.7 I would find the favourites helpful to use the pre-stored important sentences.	3.80	1.01
3.8 I liked using the interface of the favourites.	3.50	1.15
Average on reporting and customization	3.97	–
Statements for overall impression		
4.1 Overall, learning to operate EuTalk would be easy for me.	4.30	0.57
4.2 Overall, I felt comfortable using EuTalk.	3.95	0.51
4.3 Overall, I liked using the interface of EuTalk.	3.95	0.76
4.4 Overall, I could recover from errors easily and quickly using EuTalk.	3.85	0.59
4.5 Overall, I felt confident using EuTalk to communicate with others.	3.68	0.58
4.6 Overall, I am satisfied with EuTalk.	3.98	0.39
Average on overall impression	3.95	–

Table 3. The feasibility study results.

Statements for communication interface	Mean	SD
1.1 I would find the communication interface easy to use.	4.25	0.50
1.2 I was able to complete the tasks and scenarios using the communication interface.	3.50	0.58
1.3 I liked using the interface of the communication interface.	3.50	0.58
1.4 I would find the communication interface intuitive to use.	3.50	0.58
1.5 I would find the page navigation easy to understand.	4.25	0.50
1.6 I would find the function keys helpful using the communication interface.	3.75	0.96
1.7 I would find the vocabulary provided is sufficient to make daily conversation.	4.00	0.82
Average on communication interface	3.82	–
Statements for embedded training		
2.1 I would find the game easy to use.	4.00	0.00
2.2 I was able to complete the tasks using the game.	4.25	0.50
2.3 I liked using the interface of the game.	4.50	0.58
2.4 I would find the game intuitive to use.	4.25	0.50
2.5 I would find the game helpful to use the communication interface.	4.25	0.50
2.6 I would find the practice easy to use.	4.25	0.96
2.7 I was able to complete the tasks using the Practice.	4.50	0.58
2.81 I liked using the interface of the Practice.	4.25	0.50
2.91 I would find the Practice intuitive to use.	4.00	0.00
2.101 I would find the Practice helpful to use the Communication Interface.	4.00	0.82
2.11 I would find the achievement banner (practice result) motivational.	4.50	0.58
Average on Embedded Training	4.25	–
Statements for overall impression		
4.1 Overall, learning to operate EuTalk would be easy for me.	4.00	0.82
4.2 Overall, I felt comfortable using EuTalk.	4.25	0.50
4.3 Overall, I liked using the interface of EuTalk.	4.00	0.82
4.4 Overall, I could recover from errors easily and quickly using EuTalk.	4.25	0.96
4.5 Overall, I felt confident using EuTalk to communicate with others.	4.00	0.82
4.6 Overall, I am satisfied with EuTalk.	3.25	1.70
4.7 I would like to use EuTalk to communicate with others.	4.00	0.82
Average on overall impression	3.96	–

able-bodied participants made 583 utterances in total, and roughly 30 utterances per participant. Overall language performance was measured by participants' communication rates and average length of utterances. A t-test was used to evaluate participant's language performance in different demographic groups. The results show that there were no significant differences in language performance between different genders ($p=.474$), different age groups (two age groups: age <31 or age ≥ 31 , $p=.217$) and different familiarity levels of knowledge in AAC (familiar or not familiar with AAC, $p=.982$).

Discussion

The design of the EuTalk mobile app provides functional advantages over other available methods and solves the problems in those existing systems.

First, the integrated design provides not simply a communication interface, but an approach that incorporates self-learning and real-time monitoring in language rehabilitation. The selected core vocabulary in the app is able to cover daily conversations in different circumstances. The alphabet page also provides PwCD another option to enter words when they are not available in the provided vocabulary. Moreover, the customization features give PwCD and SLPs the flexibility to change the settings to meet their personal needs or adjust training materials conveniently in the app. All of these features make it possible for PwCD to achieve their full potential in language communication.

Second, the embedded interactive training modules provide PwCD an alternative way to obtain practice exercises without physical or financial barriers. This design provides PwCD with any-time and anywhere access to the training materials to fulfill the intensity requirements recommended by high quality language rehabilitation practices. The implementation of gamification makes the learning process interesting and may in turn increase user's

engagement to the app. The training can help PwCD learn how to use the app.

Third, to meet the language performance reporting requirements, from the very first time a new user account is created every single event in the app is recorded in the database. Analysis on these patient generated data is performed automatically in real time. A visual outcome report is provided to PwCD and caregivers to track the progress of the language performance. Clinicians can also access PwCD's performance at the time their patients visit their clinic and adjust the treatment accordingly.

Overall, the mobile AAC app received positive feedback from both able-bodied participants and PwCD. The minimum average on individual app component and overall impression is 3.81 out of 5. All study participants were able to finish the assigned tasks and provided satisfied ratings for the specific features of the app.

Able-bodied participants tended to have higher expectations for the accuracy of the grammar of their sentences when they used the communication interface, likely because they are used to speaking with grammatically correct sentences. The gap between their expectations for the normal language communication and their experiences with the AAC app may have led to some less favourable results. During the interview, the able-bodied participants claimed that they were very satisfied with certain features of the app, such as the embedded training materials and reporting capabilities since they could support the use of communication interface. Furthermore, participants with AAC knowledge were less concerned about generating grammar correct sentences than the rest of the participants, because they have a better understanding with the main purpose of AAC technology.

The overall satisfaction was rated relatively lower (3.25) in the feasibility study, mainly due to user's frustration. One of the PwCD participants reported feeling very tired at the end of the study because of the cognitive load required during the evaluation process, and this might be caused by the participant's

communication disabilities. All participants in the feasibility study agreed that the training materials could support the use of the communication interface and they rated the training materials higher than the able-bodied participants. The overall user experience in the usability and feasibility studies are almost the same (3.95 in the usability study vs. 3.96 in the feasibility study) in the first six statements. For the PwCD participants, there was one further statement: I would like to use EuTalk to communicate with others. This statement was not applicable to able-bodied participants. The average on the response from the PwCD participants was 4.0.

We understand the limitations of this study, such as short study period, a small number of participants with communication disability, and the results mainly from the responses of study participants. Therefore, future research is planned to conduct a clinical trial, which will evaluate the usability of the app in real clinical environment. The proposed clinical trial will include both PwCD and SLPs. A full AAC treatment will be provided to PwCD in a few months. During the study period, PwCD will ask to use this EuTalk app by following the specific instructions from SLPs every day, while SLPs will review PwCD's language performance regularly. SLPs will adjust their treatment plans according to the PwCD's performance. The data collected in this study will be used to evaluate the usability of the app quantitatively.

Conclusion

In this project, a mobile AAC app was created using a research-based approach to fill the gaps in the current language rehabilitation. A usability study and a feasibility study were conducted with two groups of participants and they were satisfied with the design and implementation. The features available in the app provide PwCD an opportunity to use the app as an AAC intervention tool conveniently. The outcome reports help PwCD to track their own progress and help SLPs to design appropriate treatment plans accordingly, which can be used to fulfill the requirements of evidence based practice.

Acknowledgments

The trial license of VoiceText™ Embedded SDK for Android platform is supported by NeoSpeech. We wish to acknowledge portions of this work were funded by a grant from the AAC Institute. We thank Semantic Compaction Systems for the reprint permission of the Pixon Project Kit.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This research is funded in part by grants from the National Institutes of Health (NIH) 1 R43 DC04246-01 and 1 R43 DC04472-01, and the National Institute on Disability, Independent Living, and Rehabilitation Research (NIDILRR) 90RE5018, 90DP0064, and 90DPGE0002. The contents of this article do not represent the views of the NIH, or NIDILRR, or the United States Government.

References

- [1] Beukelman D, Mirenda P. Augmentative and alternative communication, 2005.
- [2] Brault MW. Americans with disabilities: 2010, Current population reports, 7, 2012, pp. 1–131.
- [3] Denavas-Walt C, Proctor BD, Smith JC. US Census Bureau, current population reports, Current Population Reports, 2010.
- [4] Hill K, Baker B, Romich BA. "Augmentative and alternative communication technology". In Cooper RA, Ohnabe H, Hobson DA. An introduction to rehabilitation engineering. Abingdon: Taylor & Francis; 2007.
- [5] Donald N. The design of everyday things. New York (NY): Basic Books; 1988.
- [6] Hill K. Advances in augmentative and alternative communication as quality-of-life technology. *Phys Med and Rehabil Clin N Am*. 2010;21:43–58.
- [7] Balandin S, Iacono T. Crews, wusses, and whoppas: core and fringe vocabularies of Australian meal-break conversations in the workplace. *Augment Altern Commun*. 1999;15:95–109.
- [8] Bhogal SK, Teasell R, Speechley M. Intensity of aphasia therapy, impact on recovery. *Stroke*. 2003;34:987–993.
- [9] Sallows GO, Graupner TD. Intensive behavioral treatment for children with autism: four-year outcome and predictors. *Am J Ment Retard*. 2005;110:417–438.
- [10] Butler DL, Winne PH. Feedback and self-regulated learning: a theoretical synthesis. *Rev Edu Res*. 1995;65: 245–281.
- [11] Hill K. The development of a model for automated performance measurement and the establishment of performance indices for augmented communicators under two sampling conditions. Pittsburgh (PA): University of Pittsburgh; 2001.
- [12] Hill K. Augmentative and alternative communication and language: evidence-based practice and language activity monitoring. *Top Lang Disord*. 2004;24:18–30.
- [13] Speak For Yourself AAC, [cited Oct 2016] Available from: <http://www.speakforyourself.org/>.
- [14] Kumar S, Wang E-H, Pokabla MJ, et al. Teleophthalmology assessment of diabetic retinopathy fundus images: smartphone versus standard office computer workstation. *Telemed J E Health*. 2012;18:158–162.
- [15] Takao H, Murayama Y, Ishibashi T, et al. A new support system using a mobile device (smartphone) for diagnostic image display and treatment of stroke. *Stroke*. 2012;43:236–239.
- [16] Logan AG, McIsaac WJ, Tisler AT, et al. Mobile phone-based remote patient monitoring system for management of hypertension in diabetic patients. *Am J Hypertens*. 2007;20: 942–948.
- [17] Mattila E, Parkka J, Hermersdorf M, et al. Mobile diary for wellness management—results on usage and usability in two user studies. *IEEE Trans Inform Technol Biomed*. 2008;12:501–512.
- [18] Tomlinson M, Solomon W, Singh Y, et al. The use of mobile phones as a data collection tool: a report from a household survey in South Africa. *BMC Med Inform Decis Mak*. 2009;9:51.
- [19] Yu P, De Courten M, Pan E, et al. The development and evaluation of a Pda-based method for public health surveillance data collection in developing countries. *Int J Med Inform*. 2009;78:532–542.
- [20] Gerber BS, Stolley MR, Thompson AL, et al. Mobile phone text messaging to promote healthy behaviors and weight loss maintenance: a feasibility study. *Health Informatics J*. 2009;15:17–25.

- [21] Jordan ET, Ray EM, Johnson P, et al. Text4Baby: using text messaging to improve maternal and newborn health. *Nurs Womens Health*. 2011;15:206–212.
- [22] Chen S-HK, Wang E-H. “Developing a bilingual communication aid for reducing language barriers in the healthcare system”, *health disparities research at the intersection of race, ethnicity, and disability: a national conference*, 2013.
- [23] Wang E-H, Chen S-HK, Zhou L, et al. “EuTalk™: A Virtual Therapist and Speech Assistant for People with Communication Disabilities”, *Disability Studies Conference*, 2013.
- [24] Chen S-HK, Hill K. “A focus group and survey to inform development of a graphic symbol based Chinese AAC system”, Presented in the Annual ASHA convention. San Diego, CA, 2011.
- [25] The Pixon Project Kit, [cited Jun. 2016]. Available from: <https://store.prentrom.com/the-pixon-sup-tm-sup-project-kit>.
- [26] Deterding S, Sicart M, Nacke L, et al. *Gamification: using game-design elements in non-gaming contexts*. New York (NY): ACM Press; 2011.
- [27] Mayer RE, Moreno R. Animation as an aid to multimedia learning. *Educ Psychol Rev*. 2002;14:87–99.
- [28] Wang E-H. A m-Health platform for supporting clinical data integration and service delivery: an example from augmentative and alternative communication intervention. Pittsburgh (PA): University of Pittsburgh; 2016.
- [29] Lewis JR. IBM computer usability satisfaction questionnaires: psychometric evaluation and instructions for use. *Int J Hum-Comput Int*. 1995;7:57–78.

Copyright of Disability & Rehabilitation: Assistive Technology is the property of Taylor & Francis Ltd and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.